

Week 9 Report: Testing & Debugging

Objective

The objective of Week 9 was to test and debug all features of the Word Scramble Game. The main focus was to ensure that the game works correctly, fix logical errors, verify score calculations, test word scrambling and answer validation, and improve the readability and reliability of the Java source code.

Activities Completed

1. Testing Every Game Feature

All major game features were tested, including the welcome screen, instructions, game start, word scrambling, answer input, score system, timer, question counter, and final game summary.

2. Fixing Logical Errors

Logical errors in the game were identified and fixed. The game now handles empty answers, wrong answers, question progression, timer expiration, and game completion properly.

3. Testing Score Calculations

The score calculation was tested to ensure that the player receives 10 points for every correct answer. Wrong answers do not increase the score.

4. Verifying Scrambling and Validation

The word scrambling function was tested to ensure that the displayed word is different from the original word. Answer validation was also tested using uppercase, lowercase, and extra spaces.

5. Testing Timer Functionality

The countdown timer was tested to ensure that it decreases every second and automatically ends the game when the time reaches zero.

6. Testing Game Summary

The final game summary was tested to verify that total questions, correct answers, wrong answers, final score, and high score are displayed correctly.

7. Improving Code Readability

The Java source code was organized into separate methods for the welcome screen, instructions, game start, timer, word scrambling, answer checking, and game ending. This makes the code easier to understand and maintain.

Testing Table

Test Case	Expected Result	Status
Start Game	Game interface opens	Passed
Instructions	Instructions are displayed	Passed
Correct Answer	Score increases by 10	Passed
Wrong Answer	Score remains unchanged	Passed
Empty Answer	Warning message appears	Passed
Word Scrambling	Scrambled word differs from original	Passed
Answer Validation	Correct answer is accepted	Passed
Timer	Time decreases every second	Passed
Time Expiration	Game ends at 0 seconds	Passed
Question Counter	Question number updates correctly	Passed
Game Summary	Final statistics are displayed	Passed
Play Again	New game starts	Passed

Deliverables

- Bug Fix Report
- Updated Java Source Code
- Tested Word Scrambling System
- Tested Answer Validation
- Tested Score Calculation
- Tested Countdown Timer
- Tested Question Counter
- Final Game Summary
- Improved and Readable Java Code

Expected Outcome

The Word Scramble Game runs correctly without major logical errors. All important features including word scrambling, answer validation, score calculation, timer, question progression, and game summary work properly. The updated code is also more readable and easier to maintain.